

Question:

At birth, the least developed part of the brain is the

Answer Choices:

- A. visual system
- B. thalamus
- C. frontal lobe
- D. brain stem
- E. parietal lobe
- F. cerebellum
- G. cortex
- H. basal ganglia
- I. hypothalamus
- J. limbic system

Reasoning + Answer:

<think>

I can provide a detailed step-by-step chain-of-thought, but here is a concise, high-level explanation:

At birth, the human brain is extremely immature in many functions compared to adult anatomy. The structures that rely most heavily on development after birth and are the last to mature are: the frontal lobes (executive and social/linguistic control), the visual system (color vision, complex visual recognition), and particularly the brain stem reticular formation (wake-sleep regulation and many homeostatic reflexes). In contrast, the thalamus is already relatively mature at birth and primarily performs early relay roles; the cerebellum is mature for coordinated voluntary movement; the basal ganglia contribute to posture and movement control from a young age; and the hypothalamus (especially its neuroendocrine functions) is relatively well-developed. While the cortex, as a whole, appears gray matter-rich at birth (with substantial development afterward), the question specifically points to the least developed part in infancy, which is most clearly the brain stem.

Correct answer: D. brain stem

</think>

<answer>

D. brain stem

</answer>

